

Source-Side Action Automation by Resolution Geometry

Effective Observation Actions from Unresolved-Field Elimination

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Abstract

This paper defines a source-side automation procedure for metric-dependent field theories with finite observational resolution. Given a field action, a supplied background metric, and a resolved/unresolved split of field degrees of freedom, the construction eliminates the unresolved variables and obtains an effective observation action

$$W_{\text{obs}}[\phi_R, g; \mathcal{R}].$$

The source response is then generated by metric variation of this action, rather than by assigning resolved or unresolved observables directly to stress-energy components. For quadratic scalar actions the construction is explicit and reduces to a Schur-complement effective operator. For convex nonlinear scalar actions the same object is constructed by solving the unresolved Euler–Lagrange equations at fixed resolved field. The paper proves the eliminated-action identity, the metric-variation envelope identity, and the resulting source-side automation rule. Numerical tests on local-metric scalar lattices verify eliminated-action recovery, local metric-variation agreement, null-coupling reduction, hidden-coordinate quotient invariance, and composability of Gaussian elimination. The construction is a source-side coarse-graining mechanism; it does not modify or derive the geometric side of general relativity.

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1 Purpose

Let a physical model supply a metric-dependent field action

$$S[\phi; g].$$

A finite observation does not necessarily retain all field degrees of freedom. We write

$$\phi = (\phi_R, \phi_N),$$

where ϕ_R is resolved and ϕ_N is unresolved. The purpose of this paper is to define the effective source object seen by the resolved observer by eliminating ϕ_N at the action level:

$$(S[\phi; g], \mathcal{R}) \longrightarrow W_{\text{obs}}[\phi_R, g; \mathcal{R}] \longrightarrow T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.$$

The construction concerns the source side of a gravitational equation. It assumes that the background geometry or metric-dependent field operator has already been specified.

Non-claim 1.1 (No geometric-side derivation). This paper does not derive the Einstein tensor, the Einstein–Hilbert action, or the spacetime metric. It constructs an effective source action once a metric-dependent field model and resolution split have been provided.

Non-claim 1.2 (No manual stress dictionary). The procedure does not declare that a particular resolved scalar is energy density, pressure, flux, or anisotropic stress. The accepted source response is obtained from variation of W_{obs} .

Non-claim 1.3 (No general interacting-field theorem). The rigorous statements below cover the finite-dimensional action-level elimination setting under stated regularity and invertibility/-convexity assumptions. The numerical nonlinear example is a convex scalar test, not a theorem for arbitrary nonconvex or quantum field theories.

2 Resolution data

Definition 2.1 (Resolution package). In this paper a resolution package is

$$\mathcal{R} = (R, N, P_R, P_N, [\mathcal{O}]),$$

where R and N index the resolved and unresolved field degrees of freedom, P_R, P_N are the corresponding projectors, and $[\mathcal{O}]$ is the quotient class under admissible reparametrizations of unresolved coordinates.

The split is not interpreted as a new physical law. It specifies which degrees of freedom are retained in the resolved description and which are eliminated when constructing the effective observation action.

3 Action-level elimination

Definition 3.1 (Classical observation action). Let $S[\phi_R, \phi_N; g]$ be differentiable in ϕ_N . Suppose that for the tested (ϕ_R, g) there exists an eliminated unresolved field $\phi_N^*(\phi_R, g)$ satisfying

$$\partial_{\phi_N} S[\phi_R, \phi_N^*; g] = 0.$$

The classical observation action is

$$W_{\text{obs}}^{\text{cl}}[\phi_R, g; \mathcal{R}] = S[\phi_R, \phi_N^*(\phi_R, g); g].$$

If the unresolved sector is convex, ϕ_N^* is the unique minimizer,

$$\phi_N^* = \arg \min_{\phi_N} S[\phi_R, \phi_N; g].$$

Assumption 3.2 (Regular eliminated branch). The eliminated solution $\phi_N^*(\phi_R, g)$ exists on the tested neighbourhood and either is unique or belongs to a declared smooth branch. At the eliminated point the unresolved Hessian is nonsingular:

$$\partial_{\phi_N \phi_N}^2 S[\phi_R, \phi_N^*; g] \text{ is invertible.}$$

Theorem 3.3 (Eliminated-action identity). *Under the preceding definition,*

$$W_{\text{obs}}^{\text{cl}}[\phi_R, g; \mathcal{R}] = S[\phi_R, \phi_N^*; g].$$

This identity is definitional for general stationary elimination and becomes an explicit computable formula once ϕ_N^ is constructed.*

Proof. The observation action is defined by evaluating the supplied action on the eliminated unresolved solution. The content of the theorem is not a new algebraic identity but the operational rule: the effective resolved action is obtained by solving the unresolved Euler–Lagrange equations at fixed ϕ_R and substituting the solution back into S . $\square$

Theorem 3.4 (Metric-variation envelope identity). *Let $g \mapsto S[\phi_R, \phi_N; g]$ and $g \mapsto \phi_N^*(\phi_R, g)$ be differentiable on a regular eliminated branch. For any metric parameter λ ,*

$$\frac{\partial W_{\text{obs}}^{\text{cl}}}{\partial \lambda} = \left. \frac{\partial S[\phi_R, \phi_N; g]}{\partial \lambda} \right|_{\phi_N = \phi_N^*}.$$

Proof. By the chain rule,

$$\frac{d}{d\lambda} S[\phi_R, \phi_N^*(\lambda); g(\lambda)] = \frac{\partial S}{\partial \lambda} + \left(\frac{\partial S}{\partial \phi_N} \right)^T \frac{\partial \phi_N^*}{\partial \lambda}.$$

At the eliminated solution, $\partial S / \partial \phi_N = 0$. Therefore the dependence of the eliminated hidden solution does not contribute to first order, and only the explicit metric derivative remains. $\square$

Corollary 3.5 (Automated source response). *On a regular eliminated branch, the source response of the resolved description is generated by varying W_{obs} :*

$$T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.$$

For finite-dimensional metric tests this is represented by derivatives $\partial W_{\text{obs}} / \partial \lambda_i$. The source response is therefore obtained from action elimination, not from manual assignment of hidden variables to tensor components.

4 Quadratic scalar sector

For a quadratic finite-dimensional scalar action,

$$S[\phi; g] = \frac{1}{2} \phi^T A[g] \phi,$$

write

$$A[g] = \begin{pmatrix} A_{RR} & A_{RN} \\ A_{NR} & A_{NN} \end{pmatrix}.$$

If A_{NN} is invertible, the unresolved equation is

$$A_{NR} \phi_R + A_{NN} \phi_N = 0,$$

so

$$\boxed{\phi_N^* = -A_{NN}^{-1} A_{NR} \phi_R.}$$

Substitution gives

$$\boxed{W_{\text{obs}}^{\text{cl}}[\phi_R, g; \mathcal{R}] = \frac{1}{2} \phi_R^T \left(A_{RR} - A_{RN} A_{NN}^{-1} A_{NR} \right) \phi_R.}$$

The effective resolved operator is

$$A_{\text{eff}} = A_{RR} - A_{RN} A_{NN}^{-1} A_{NR}.$$

Proposition 4.1 (Schur construction). *For quadratic actions with invertible A_{NN} , action-level unresolved-field elimination is exactly the Schur complement of the unresolved block.*

Proof. Substitute $\phi_N^* = -A_{NN}^{-1} A_{NR} \phi_R$ into $S = \frac{1}{2} \phi^T A \phi$. The remaining quadratic form on ϕ_R is precisely the Schur complement displayed above. $\square$

5 Numerical protocols

The tests in this paper use only gates required by the construction.

Protocol 5.1 (Stationary elimination). Construct ϕ_N^* either by solving the linear Schur equation or by numerical minimization/stationary solve. Report the residual

$$\|\partial_{\phi_N} S\|_{\infty}.$$

Protocol 5.2 (Metric-variation agreement). For local metric probes θ_p , compare finite differences of the eliminated action,

$$\frac{W_{\text{obs}}(\theta_p + h) - W_{\text{obs}}(\theta_p - h)}{2h},$$

with the explicit derivative

$$\left. \frac{\partial S}{\partial \theta_p} \right|_{\phi_N = \phi_N^*}.$$

Protocol 5.3 (Null-coupling reduction). Remove the resolved/unresolved coupling. In the quadratic case this means $A_{RN} = A_{NR} = 0$. In the nonlinear finite-dimensional test it means removing the cross-coupling induced by the lattice kinetic operator. The eliminated hidden solution should reduce to the hidden-sector vacuum for the chosen potential.

Protocol 5.4 (Hidden-coordinate quotient invariance). Reparametrize the hidden coordinate by $y = s\phi_N$. The construction must be invariant when the physical field is recovered as $\phi_N = y/s$. Raw use of y as if it were the physical hidden field is not quotient-invariant.

Protocol 5.5 (Composable elimination). In the Gaussian case, compare direct elimination of two unresolved blocks with stepwise elimination. Agreement tests whether the effective action is independent of the elimination order up to numerical tolerance.

6 Numerical model I: local-metric Gaussian scalar

The Gaussian test uses a periodic $N_t \times N_x$ lattice and a local conformal metric-like field

$$\theta(t, x) = 0.18 \cos \frac{2\pi x}{N_x} + 0.07 \sin \left(\frac{2\pi t}{N_t} + 0.3 \right).$$

The Euclidean operator is

$$A_E = L_t + L_x + m^2 \text{diag}(e^{2\theta}),$$

and the Lorentzian classical saddle operator is

$$A_L = -L_t + L_x + m^2 \text{diag}(e^{2\theta}).$$

The resolved/unresolved split is checkerboard:

$$R = \{(t, x) : t + x \equiv 0 \pmod{2}\}, \quad N = \{(t, x) : t + x \equiv 1 \pmod{2}\}.$$

6.1 Gaussian Euclidean results

Grid	cond(A_{NN})	correction norm	eliminated-action error	local-gradient error	quotient error
8×8	1.1589	0.2690	3.55×10^{-15}	1.78×10^{-9}	0
12×12	1.1626	0.2690	7.11×10^{-15}	7.11×10^{-9}	0
16×16	1.1647	0.2690	7.11×10^{-15}	3.55×10^{-9}	7.11×10^{-15}

Finding 6.1 (Gaussian Euclidean source action). The Schur observation action recovers the eliminated action to roundoff and its local metric variation agrees with the full eliminated action variation to finite-difference precision.

6.2 Gaussian Lorentzian saddle results

Grid	negative modes in A_L	negative modes in A_{eff}	cond(A_{NN})	correction norm	eliminated-action error	local-gradient error
8×8	21	21	2.6318	12.1650	1.33×10^{-15}	8.88×10^{-10}
12×12	53	53	2.6845	12.2072	3.55×10^{-15}	8.88×10^{-10}
16×16	89	89	2.7150	12.2233	0	3.55×10^{-9}

Finding 6.2 (Gaussian Lorentzian saddle action). The Lorentzian-signature Gaussian action is indefinite, but the unresolved block remains invertible in the tested charts. Classical saddle elimination and local metric variation agreement remain numerically stable.

Warning 6.3 (Lorentzian determinant). The Lorentzian table concerns classical saddle elimination. It does not define a positive measure and does not solve the phase/sign problem of Lorentzian quantum functional integration.

6.3 Gaussian composability

Sector	8×8	12×12	16×16
Euclidean	6.82×10^{-17}	9.21×10^{-17}	9.43×10^{-17}
Lorentzian	5.85×10^{-16}	1.31×10^{-15}	2.01×10^{-15}

Finding 6.4 (Composable Gaussian elimination). Direct and stepwise Schur elimination agree to numerical precision in the tested Gaussian models.

7 Numerical model II: actual nonlinear elimination

The nonlinear test uses the Euclidean scalar action

$$S[\phi; \theta] = \frac{1}{2} \phi^T (L_t + L_x) \phi + \sum_p e^{2\theta_p} \left(\frac{1}{2} m^2 \phi_p^2 + \frac{1}{4} \lambda \phi_p^4 \right),$$

with $m^2 = 0.8$ and $\lambda = 0.18$. The observation action is constructed directly by optimization:

$$W_{\text{obs}}[\phi_R, \theta] = \min_{\phi_N} S[\phi_R, \phi_N; \theta].$$

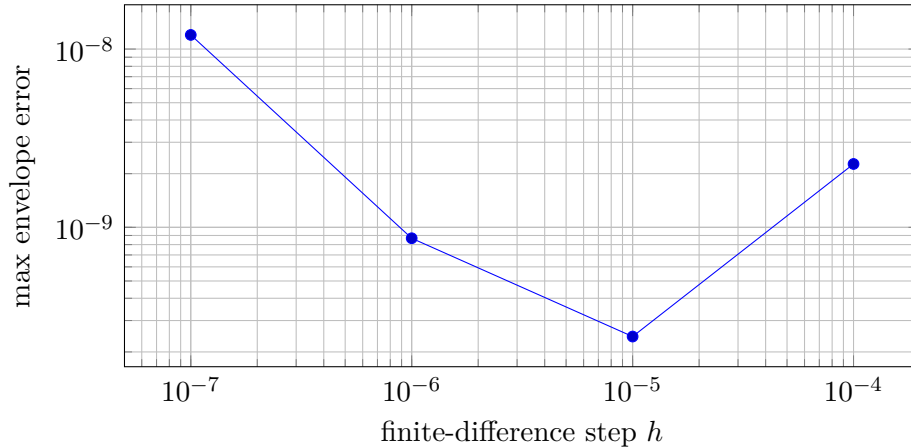
Grid	W_{obs}	stationarity residual	envelope error	quotient error	null error
6×6	6.1570	6.24×10^{-9}	1.16×10^{-9}	0	0
8×8	8.7914	8.68×10^{-10}	1.07×10^{-9}	0	0
10×10	11.5419	1.75×10^{-9}	8.68×10^{-10}	0	0
12×12	14.6439	4.19×10^{-9}	8.46×10^{-10}	0	0

Finding 7.1 (Actual unresolved-field elimination). In the convex nonlinear scalar test, W_{obs} is constructed by directly eliminating ϕ_N . The eliminated solution is stationary to approximately 10^{-9} in the refined optimizer check, and the local metric envelope identity holds to approximately 10^{-9} .

For hidden-coordinate scaling $y = 3\phi_N$, W_{obs} remains invariant at displayed precision. A raw hidden-coordinate manual energy is not invariant because the physical field is $y/3$, not y . With the quartic term its scaling is not a pure factor of nine; the observed ratios lie between approximately 9.34 and 10.12 on the tested grids.

7.1 Finite-difference stability for nonlinear elimination

On the 10×10 nonlinear model the local metric envelope error behaves as follows:



The error is smallest near $h = 10^{-5}$ and worsens for very small h , consistent with finite-difference roundoff.

8 Reader protocol

A reader can reproduce the construction as follows.

1. Choose a metric-dependent scalar action $S[\phi; g]$.
2. Choose a finite resolution split $\phi = (\phi_R, \phi_N)$.
3. At fixed ϕ_R , solve the unresolved equation

$$\partial_{\phi_N} S = 0.$$

For convex Euclidean sectors solve it as a minimization problem.

4. Define

$$W_{\text{obs}}[\phi_R, g] = S[\phi_R, \phi_N^*; g].$$

5. Verify stationarity of the eliminated hidden solution.
6. Verify the local metric envelope identity by finite differences.
7. Check quotient invariance by reparametrizing hidden coordinates and recovering the same physical field.
8. In Gaussian sectors, compare direct and stepwise Schur elimination.

9 Conclusion

This paper gives a finite-resolution source-side construction:

$$S[\phi; g], \mathcal{R} \longrightarrow \phi_N^* \longrightarrow W_{\text{obs}}[\phi_R, g; \mathcal{R}] \longrightarrow T_{\mu\nu}^{\text{obs}}.$$

For quadratic scalar actions the observation action is the Schur complement of the unresolved block. For convex nonlinear scalar actions it is obtained by actual unresolved-field elimination. The metric-variation envelope identity explains why the effective source response is generated by varying W_{obs} , without assigning hidden variables directly to stress components. The numerical tests verify the construction on local-metric scalar lattices.

The construction is therefore a source-side action automation pipeline. It does not attempt to derive the gravitational geometry; it automates how a finite-resolution field description produces an effective source action and its metric response.

A Background demonstration: FRW and Schwarzschild source automation

This appendix applies the same construction to two supplied backgrounds. The purpose is not to solve the metric equations. The metric is fixed, the scalar action is discretized on that background, the unresolved field values are eliminated, and the effective observation action is differentiated with respect to local background weights.

A.1 FRW conformal scalar model

For the FRW demonstration we use a spatially periodic 1 + 1-dimensional conformal model

$$ds^2 = a(\eta)^2(-d\eta^2 + dx^2).$$

After Euclideanization for convex numerical elimination, the conformal kinetic part is represented by unit nearest-neighbour weights and the local potential weight is

$$V_{\text{FRW}}(\eta, x) = a(\eta)^2.$$

The scale factor used in the battery is

$$a(\eta) = 1 + 0.22 \sin \frac{2\pi\eta}{N_\eta} + 0.08 \cos \frac{4\pi\eta}{N_\eta}.$$

The nonlinear scalar action is

$$S[\phi; a] = \frac{1}{2} \sum_{\langle pq \rangle} (\phi_p - \phi_q)^2 + \sum_p a_p^2 \left(\frac{1}{2} m^2 \phi_p^2 + \frac{1}{4} \lambda \phi_p^4 \right),$$

with $m^2 = 0.7$ and $\lambda = 0.12$. The observation action is constructed by

$$W_{\text{obs}}[\phi_R, a] = \min_{\phi_N} S[\phi_R, \phi_N; a].$$

Grid	W_{obs}	stationarity	relative stationarity	envelope error	envelope rel.	quotient error
8×8	6.1238	7.37×10^{-9}	2.34×10^{-9}	2.92×10^{-10}	1.15×10^{-9}	0
10×10	7.9233	1.11×10^{-9}	3.83×10^{-10}	7.36×10^{-11}	2.89×10^{-10}	0
12×12	9.9286	1.03×10^{-9}	3.84×10^{-10}	8.07×10^{-11}	3.17×10^{-10}	0

Finding A.1 (FRW source-side automation). For the tested FRW conformal scalar model, unresolved-field elimination succeeds, the eliminated solution is stationary to approximately 10^{-9} , local background-weight variation satisfies the envelope identity to approximately 10^{-10} – 10^{-9} , quotient reparametrization gives no displayed change in W_{obs} , and null-coupling reduction gives zero hidden norm and zero action error.

A.2 Schwarzschild exterior radial model

For the Schwarzschild demonstration we use the exterior static metric through its radial/tortoise weights. Let

$$f(r) = 1 - \frac{2M}{r}, \quad r_* = r + 2M \log \left(\frac{r}{2M} - 1 \right),$$

with $M = 1$ and $r \in [6, 20]$, so the test remains outside the horizon. A spherically reduced Euclideanized scalar action is represented by shell-area kinetic weights r^2 and potential weights

$$V_{\text{Schw}}(r) = r^2 f(r).$$

Thus the discrete action has the form

$$S[\phi; r] = \frac{1}{2} \sum_{\langle pq \rangle} w_{pq} (\phi_p - \phi_q)^2 + \sum_p r_p^2 f(r_p) \left(\frac{1}{2} m^2 \phi_p^2 + \frac{1}{4} \lambda \phi_p^4 \right).$$

The observation action is again

$$W_{\text{obs}}[\phi_R, r] = \min_{\phi_N} S[\phi_R, \phi_N; r].$$

Grid	W_{obs}	stationarity	relative stationarity	envelope error	envelope rel.	quotient error
8×8	988.5031	7.48×10^{-5}	1.37×10^{-7}	8.28×10^{-7}	2.50×10^{-8}	6.59×10^{-12}
10×10	1274.5282	6.95×10^{-5}	1.64×10^{-7}	9.20×10^{-7}	2.47×10^{-8}	4.32×10^{-12}
12×12	1590.1199	5.29×10^{-5}	1.18×10^{-7}	5.51×10^{-7}	1.12×10^{-8}	4.32×10^{-12}

Finding A.2 (Schwarzschild exterior source-side automation). For the tested exterior Schwarzschild radial model, unresolved-field elimination succeeds on all grids. Because the weights are much larger than in the normalized FRW case, the absolute hidden stationarity residual is around 10^{-5} , but its relative residual is around 10^{-7} . The local envelope identity holds at relative error 10^{-8} , quotient reparametrization changes W_{obs} only at roundoff-level, and null-coupling reduction gives zero hidden norm and zero action error.

Warning A.3 (Scope of the Schwarzschild battery). The Schwarzschild battery is an exterior, radial, Euclideanized source-construction test. It demonstrates the action-elimination pipeline on a supplied Schwarzschild background. It does not evolve the Schwarzschild metric, impose gravitational constraints, cross the horizon, or compute a full four-dimensional stress tensor.

A.3 Background battery protocol

Both demonstrations use the same operational protocol.

1. Build the weighted scalar action from the supplied background.
2. Choose a checkerboard resolved/unresolved split.
3. Solve $\partial_{\phi_N} S = 0$ by convex minimization at fixed ϕ_R .
4. Set $W_{\text{obs}} = S[\phi_R, \phi_N^*; g]$.
5. Check stationarity of ϕ_N^* .
6. Perturb local background potential weights and compare finite differences of W_{obs} with the explicit envelope derivative.
7. Reparametrize hidden variables by $y = 3\phi_N$ and verify invariance of W_{obs} .
8. Remove resolved/unresolved kinetic couplings and verify hidden vacuum/null reduction.

A.4 Minimal background-battery Python entry point

The executable battery accompanying this appendix is saved as `rg_schwarzschild_frw_automation_battery.py`. Its main entry point is:

```
if __name__ == '__main__':
    for bg in ['FRW', 'Schwarzschild']:
        for N in [8, 10, 12]:
            print(run_case(bg, N, N))
```

The function `run_case` constructs the background weights, eliminates ϕ_N , evaluates W_{obs} , checks stationarity, checks the local envelope identity, checks hidden-coordinate quotient invariance, and checks null-coupling reduction.

B Minimal Python implementation: nonlinear elimination

```
import numpy as np
from scipy.optimize import minimize

def idx(t, x, Nt, Nx): return t * Nx + x

def build_laplacians(Nt, Nx):
    N = Nt * Nx
    Lt = np.zeros((N, N)); Lx = np.zeros((N, N))
```

```

    for t in range(Nt):
        for x in range(Nx):
            p = idx(t, x, Nt, Nx)
            Lt[p, p] += 2
            Lt[p, idx((t-1) % Nt, x, Nt, Nx)] -= 1
            Lt[p, idx((t+1) % Nt, x, Nt, Nx)] -= 1
            Lx[p, p] += 2
            Lx[p, idx(t, (x-1) % Nx, Nt, Nx)] -= 1
            Lx[p, idx(t, (x+1) % Nx, Nt, Nx)] -= 1
    return Lt, Lx

def split_checker(Nt, Nx):
    R, N = [], []
    for t in range(Nt):
        for x in range(Nx):
            (R if (t+x) % 2 == 0 else N).append(idx(t, x, Nt, Nx))
    return np.array(R), np.array(N)

def action(phi, theta, K, m2=0.8, lam=0.18):
    vol = np.exp(2 * theta.reshape(-1))
    return 0.5 * float(phi @ K @ phi) + float(np.sum(vol * (0.5*m2*phi**2 + 0.25*lam*phi**4)))

def eliminate_phiN(r, R, N, theta, K, m2=0.8, lam=0.18):
    vol = np.exp(2 * theta.reshape(-1))
    def fun(n):
        phi = np.zeros(theta.size); phi[R] = r; phi[N] = n
        return action(phi, theta, K, m2, lam)
    def jac(n):
        phi = np.zeros(theta.size); phi[R] = r; phi[N] = n
        grad = K @ phi + vol * (m2 * phi + lam * phi**3)
        return grad[N]
    res = minimize(fun, np.zeros(len(N)), jac=jac, method="L-BFGS-B",
        options={"gtol": 1e-12, "ftol": 1e-14, "maxiter": 2000})
    return res.fun, res.x, res

Nt = Nx = 10
Lt, Lx = build_laplacians(Nt, Nx)
K = Lt + Lx
T, X = np.meshgrid(np.arange(Nt), np.arange(Nx), indexing="ij")
theta = 0.16*np.cos(2*np.pi*X/Nx) + 0.05*np.sin(2*np.pi*T/Nt + 0.3)
R, N = split_checker(Nt, Nx)
r = np.array([0.55*np.cos(2*np.pi*(p%Nx)/Nx) - 2*np.pi*(p//Nx)/Nt)
    + 0.18*np.sin(4*np.pi*(p%Nx)/Nx + 0.7) for p in R])
Wobs, nstar, res = eliminate_phiN(r, R, N, theta, K)
print("Wobs", Wobs)
print("success", res.success)
print("stationarity", np.linalg.norm(res.jac, np.inf))

```

References

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